

Dart Programming Lab 1

Screenshots

Part I: Functions

Q1.

```
C:\Users\USER\Documents\dart-school-lab\my_first_flutter_app>dart run Dart-Lab\Part1\welcome.dart
Welcome to the School Management System!
```

Q2.

```
C:\Users\USER\Documents\dart-school-lab\my_first_flutter_app>dart run Dart-Lab\Part1\createStudent.dart
Student Name: Alice
Student Age: 20

C:\Users\USER\Documents\dart-school-lab\my_first_flutter_app>dart run Dart-Lab\Part1\createStudent.dart
Student Name: Patrick
Student Age: 24
```

Q3.

```
my_first_flutter_app / Dart-Lab / Part1 / createTeacher.dart / main
2 void createTeacher(String name, [String? subject]) {
4
5     if (subject != null) {
6         print("Subject: $subject");
7     } else {
8         print("Subject not assigned");
9     }
10 }
11 //Calling Function with and without optional parameter
Run | Debug | Profile
12 void main() {
13     createTeacher("Mr. Vincent", "Dart Programming");
14     createTeacher("Dr. Patrick");
15 }

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SONARQUBE SPELL CHECKER JavaSE-2

C:\Users\USER\Documents\dart-school-lab\my_first_flutter_app>dart run Dart-Lab\Part1\createTeacher.dart
Teacher Name: Mr. Vincent
Subject: Dart Programming
Teacher Name: Dr. Patrick
Subject not assigned
```

Part II: Constructors and Classes

Q4.

```
my_first_flutter_app > Dart-Lab > Part2 > Student.dart > ...
1 //class to represent a Student
2 class Student {
3     String name;
4     int age;
5
6     // Constructor
7     Student(this.name, this.age);
8 }
```

Q5.

```
1 class Student {
2     String name;
3     int age;
4
5     Student(this.name, this.age);
6 }
7 // Using the Student class
Run | Debug | Profile
8 void main() {
9     Student student1 = Student("Patrick", 24);
10
11     print("Student Name: ${student1.name}");
12     print("Student Age: ${student1.age}");
13 }

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SONARQUBE SPELL CHECKER

C:\Users\USER\Documents\dart-school-lab\my_first_flutter_app>dart run Dart-Lab\Part2\studentObject.dart
Student Name: Patrick
Student Age: 24
```

Part III: Inheritance

Q6.

```
1 //class to represent a person
2
3 class Person {
4   String name;
5
6   Person(this.name);
7
8   void introduce() {
9     print("Hello, my name is $name.");
10  }
11 }
```

Q7.

```
11 // Student class inheriting from Person
12 class Student extends Person {
13   int age;
14
15   Student(String name, this.age) : super(name);
16 }
17
18 Run | Debug | Profile
19 void main() {
20   Student student1 = Student("Patrick", 24);
21   student1.introduce();
22 }
23
24 PROBLEMS 18 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SONARQUBE SPELL CHECKER 2
25
26 C:\Users\USER\Documents\dart-school-lab\my_first_flutter_app>dart run Dart-Lab\Part3\Inheritance.dart
27 Hello, my name is Patrick.
```

Part IV: Interfaces

Q8.

```
1 abstract class Registrable {
2   void registerCourse(String courseName);
3 }
4
```

Q9.

```
15 class Student extends Person implements Registrable {
16   int age;
17
18   Student(String name, this.age) : super(name);
19
20   @override
21   void registerCourse(String courseName) {
22     print("$name has registered for the course: $courseName");
23   }
24 }
25
26 Run | Debug | Profile
27 void main() {
28   Student student1 = Student("Patrick", 20);
29   student1.introduce();
30   student1.registerCourse("Mobile App Development");
31 }
32
33 PROBLEMS 23 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SONARQUBE SPELL CHECKER 2
34
35 C:\Users\USER\Documents\dart-school-lab\my_first_flutter_app>dart run Dart-Lab\Part4\studentImplement.dart
36 Hello, my name is Patrick.
37 Patrick has registered for the course: Mobile App Development
38 C:\Users\USER\Documents\dart-school-lab\my_first_flutter_app>
```

Part V: Mixins

Q10.

```
1 mixin AttendanceMixin {
2   int attendanceCount = 0;
3
4   void markAttendance() {
5     attendanceCount++;
6   }
7 }
```

Q11.

```
21 }
22
23 class Student extends Person
24     with AttendanceMixin
25     implements Registrable {
26
27     int age;
28
29     Student(String name, this.age) : super(name);
30
31     @override
32     void registerCourse(String courseName) {
33         print("$name has registered for the course: $courseName");
34     }
35 }
36
37 void main() {
38     Student student1 = Student("Patrick", 20);
39
40     student1.markAttendance();
41     student1.markAttendance();
42     student1.markAttendance();
43
44     print("Attendance count: ${student1.attendanceCount}");
45 }
46
```

Run | Debug | Profile

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SONARQUBE SPELL CHECKER

C:\Users\USER\Documents\dart-school-lab\my_first_flutter_app>dart run Dart-Lab\Part5\studentAttendanceMixin.dart
Attendance count: 3

Part VI: Collections

Q12.

```
11 void main() {
12     List<Student> students = [];
13
14     students.add(Student(1, "Patrick", 20));
15     students.add(Student(2, "Vincent", 22));
16     students.add(Student(3, "Straton", 21));
17
18     for (var student in students) {
19         print("ID: ${student.id}, Name: ${student.name}, Age: ${student.age}");
20     }
21 }
22
```

Run | Debug | Profile

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SONARQUBE SPELL CHECKER

C:\Users\USER\Documents\dart-school-lab\my_first_flutter_app>dart run Dart-Lab\Part6\studentList.dart
ID: 1, Name: Patrick, Age: 20
ID: 2, Name: Vincent, Age: 22
ID: 3, Name: Straton, Age: 21

Q13.

```
15 // Using a Map to store students with their ID as the key
16 void main() {
17     Map<int, Student> studentMap = {
18         1: Student(1, "Patrick", 20),
19         2: Student(2, "Vincent", 22),
20         3: Student(3, "Straton", 21),
21     };
22
23     // Print all student names
24     studentMap.forEach((id, student) {
25         print(student.name);
26     });
27 }
28
```

Run | Debug | Profile

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SONARQUBE SPELL CHECKER

C:\Users\USER\Documents\dart-school-lab\my_first_flutter_app>dart run Dart-Lab\Part6\studentMap.dart
Patrick
Vincent
Straton

Part VII: Anonymous arrow functions

Q14.

```
10 //List Using the Student class
11 void main() {
12     List<Student> students = [];
13
14     students.add(Student(1, "Patrick", 20));
15     students.add(Student(2, "Vincent", 22));
16     students.add(Student(3, "Straton", 21));
17     //Anonymous function to print student names
18     students.forEach((student) {
19         print(student.name);
20     });
21 }
22
```

Run | Debug | Profile

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SONARQUBE SPELL CHECKER

C:\Users\USER\Documents\dart-school-lab\my_first_flutter_app>dart run Dart-Lab\Part7\AnonymousFn.dart
Patrick
Vincent
Straton

Q15.

```
1 //define an arrow function to greet a student
2 void greetStudent(String name) => print("Hello, $name! Welcome.");
3 void main() {
4   greetStudent("Patrick");
5 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SONARQUBE SPELL CHECKER

C:\Users\USER\Documents\dart-school-lab\my_first_flutter_app>dart run Dart-Lab\Part7\arrowFunction.dart
Hello, Patrick! Welcome.

Part VIII: Asynchronous Programming

Q16.

```
2 class Student {
3   Student(this.id, this.name, this.age);
4 }
5
6 Future<List<Student>> loadStudents() async {
7   await Future.delayed(Duration(seconds: 2));
8
9   return [
10     Student(1, "Alice", 20),
11     Student(2, "Bob", 22),
12     Student(3, "Charlie", 21),
13   ];
14 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SONARQUBE SPELL CHECKER

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Q17.

```
9 Future<List<Student>> loadStudents() async {
10   await Future.delayed(Duration(seconds: 2));
11
12   return [
13     Student(1, "Alice", 20),
14     Student(2, "Bob", 22),
15     Student(3, "Charlie", 21),
16   ];
17 }
18
19 Future<void> main() async {
20   List<Student> students = await loadStudents();
21   print("Number of students loaded: ${students.length}");
22 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SONARQUBE SPELL CHECKER

JavaSE-21 LTS

C:\Users\USER\Documents\dart-school-lab\my_first_flutter_app>dart run Dart-Lab\Part8\asyncFnCall.dart
Number of students loaded: 3

Part V III: Integration Challenge

Q19.

```
1 //NotificationMixin.dart
2
3 mixin NotificationMixin {
4   void notifyRegistration(String courseName) {
5     print("Student has been registered to the course: $courseName");
6   }
7 }
8
9 class Student with NotificationMixin {
10   int id;
11   String name;
12   Student(this.id, this.name);
13 }
14
15 void main() {
16   Student student = Student(1, "Patrick");
17   student.notifyRegistration("Mobile App Development");
18 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SONARQUBE SPELL CHECKER

JavaSE-21 LTS

C:\Users\USER\Documents\dart-school-lab\my_first_flutter_app>dart run Dart-Lab\Part9\NotificationMixin.dart
Student has been registered to the course: Mobile App Development